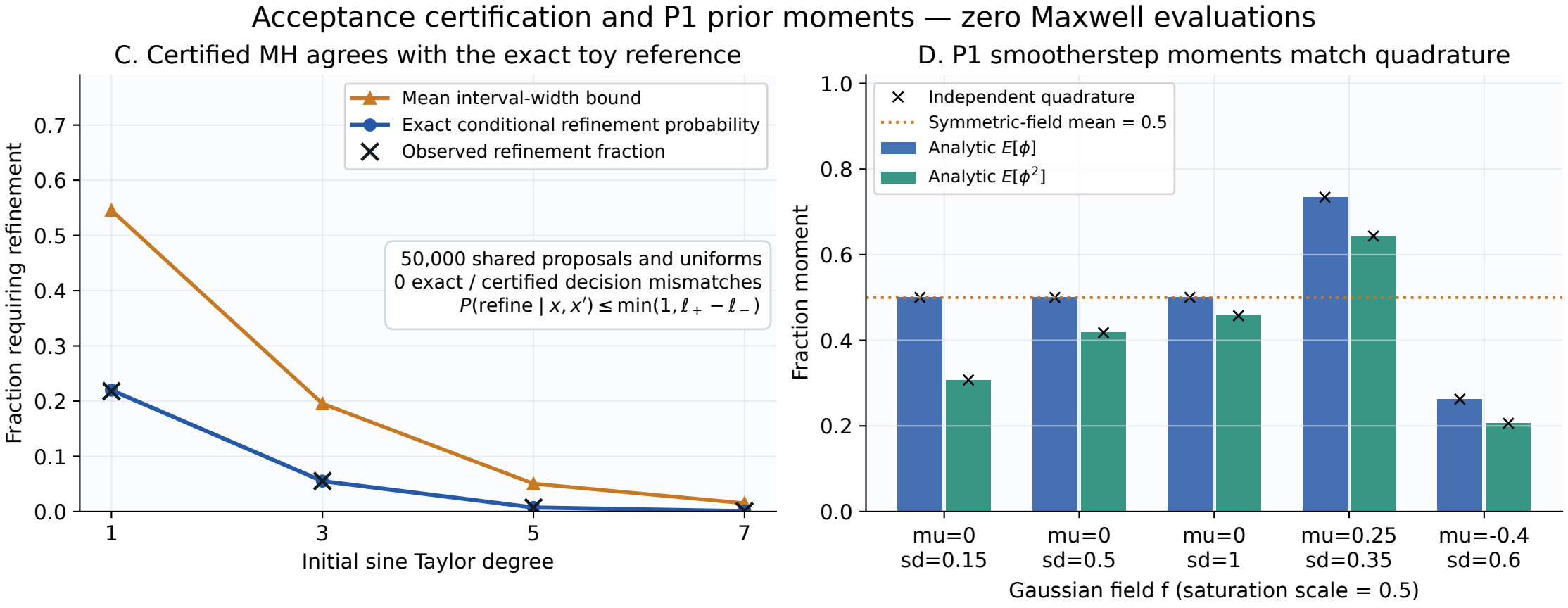
Acceptance certification and P1 prior moments — zero Maxwell evaluations
C. Certified MH agrees with the exact toy reference
Mean interval-width bound
Exact conditional refinement probability
Observed refinement fraction
50,000 shared proposals and uniforms
0 exact / certified decision mismatches
$P(\text{refine} \mid x, x') \le \min(1, \ell_+ - \ell_-)$
Fraction requiring refinement
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
1
3
5
7
Initial sine Taylor degree
D. P1 smootherstep moments match quadrature
Independent quadrature
Symmetric-field mean = 0.5
Analytic $E[\phi]$
Analytic $E[\phi^2]$
Fraction moment
0.0
0.2
0.4
0.6
0.8
1.0
mu=0
sd=0.15
mu=0
sd=0.5
mu=0
sd=1
mu=0.25
sd=0.35
mu=-0.4
sd=0.6
Gaussian field f (saturation scale = 0.5)